Dean Jie Zhong

deanjiezhong@gmail.com · +86 18915504270

Personal Website · ORCID · Google Scholar

Education

BSc (Hons) Biological Sciences, Xi'an Jiaotong-Liverpool University, Suzhou, China (2023–2027, expected)

- Expected degrees from both Xi'an Jiaotong-Liverpool University and the University of Liverpool.
- GPA: 3.74/4.00, Weighted Average: 70/100 (Transcript↗) · Expected First-Class Honours
- Relevant Modules: The Biology of Microbes 83%; Cell Biology Techniques 90%.

Research Experience

TroGen: A commensal bacterial platform for intracellular drug delivery in TNBC

iGEM 2025, School of Science, XJTLU (Feb–Oct 2025)

- Independently conceptualized and led the project; designed all genetic systems and experiments.
- Designed the *fnbA-sfgfp* bicistronic cassette and established bacteria–cell co-culture workflows across four mammalian cell lines; measured >99% invasion in MDA-MB-231 cells by flow cytometry and observed no co-localization between bacterial signals and endosomes by confocal imaging.
- Engineered a hypoxia-responsive module from the *nreABC* operon in *S. xylosus*, observing ~3-fold higher reporter fluorescence under hypoxia than normoxia.
- Analyzed RM systems; designed the HsdMS pre-methylation construct for *S. epidermidis* and selected *E. coli* JM110 for *S. xylosus* plasmid propagation, supporting the team's successful transformation of *S. xylosus*.
- Wrote scientific wiki content and prepared figures; represented the team in the project presentation and Q&A at the Grand Jamboree in Paris.

Engineering for Cancer Immunotherapy

Intern, Suzhou Institute of Systems Medicine, CAMS & PUMC (Jan 2026–present)

- **IL4–IL8 heterodimeric Fc fusion protein:** Evaluated fusion proteins through protein purification, neutrophil–B16OVA–OT-I co-culture, flow cytometry, and LDH assays; contributing to studies showing increased T cell activation and $GZMB^+$ CD8 T cell frequency.
- **CXCL13Fc fusion protein and CXCR5-engineered CD8 T cells:** Repeatedly performed viral packaging, T cell isolation, activation, transduction and expansion; independently performed flow cytometry and FlowJo analysis. Expressed and purified CXCL13Fc and controls, and performed mouse tumor inoculation, tail-vein adoptive transfer and tissue collection, supporting studies of adoptive T cell therapy.
- **TGFβ signal-switching receptor engineering:** Performed viral packaging, naïve T cell isolation and activation, transduction and culture expansion; detected no TGFβ-induced STAT6 phosphorylation.
- **Cesome exploration for plasmid endotoxin removal:** Proposed and independently designed screening of cerium-containing materials; observed LPS clearance, then designed and tested plasmid-recovery procedures and assessed DNA recovery and material compatibility using NanoDrop and DLS.

Motility Mechanisms in *Flavobacterium johnsoniae*

Undergraduate Research, Dr. Yongtao Zhu's Lab, XJTLU (Jan 2026–present)

- Conceptualized and led experimental development of a fluorescence-based approach to distinguish bacterial migration from proliferation, with ongoing refinement of staining conditions.
- Proposed an alternative labeling strategy for SprB and led antibody-labeling trials using strains expressing sfGFP–SprB fusions, assessing labeling specificity and reliability for single-particle tracking.
- Developed and iteratively tested Python workflows for temporal color projection (Rainbow Trace) and cell identity tracking (Rainbow Tracker).
- Investigated colony polysaccharide organization through iterative refinement of ConA staining protocols and hardware design.

Publications

Zhong, J.#, Wang, J.#, Wang, C.*, Zhang, L.*(2026). Genome-wide human T cell screens: barriers and design. *Trends in Cancer*. https://doi.org/10.1016/j.trecan.2026.10.004 (# Equal contribution; co-first authors.)

- Conceived the context–readout framework and drafted the full manuscript, synthesizing recent T-cell CRISPR screens to explain how functional readouts prioritize specific perturbation effects while T-cell identity and tumor microenvironments shape those effects, together informing hit interpretation, cross-screen comparisons, and screening design.

Honors & Awards

International Genetically Engineered Machine (iGEM) Competition (Oct 2025)

- Top 10 (among 190 undergrad teams); Best New Composite Part (sole undergrad winner); Gold Medal.
- Nominated for Best Oncology Project (Undergrad & Overgrad) and Best Wiki (Undergrad).

Third Prize (Jiangsu Province), China High School Biology Olympiad (Nov 2022, pre-university)

Academic Mentorship

Advisor, Zhejiang University–University of Edinburgh Institute iGEM Team 2026 (Jul 2026–present)

- Advised on project design, experimental planning, and laboratory techniques; provided strains and plasmids.

Instructor, Soochow University ICPI iGEM Team 2026 (May 2026–present)

- Founded the college's first iGEM team and secured university funding (200,000 CNY/Year); advised on project design, experimental planning, and laboratory techniques; facilitated external collaborations.

Advisor, Xi'an Jiaotong-Liverpool University iGEM Team 2026 (Jan 2026–present)

- Supported recruitment, guided project conceptualization, assisted in securing university sponsorship; advised on project development and facilitated external collaborations.

Presentations & Posters

Zhong, J. (2026). *A commensal bacterial platform for intracellular drug delivery in TNBC*, Poster, BIOCHINA Youth Research Exhibition, Suzhou International Expo Centre, Suzhou, China (14 Mar 2026)

Zhong, J. (2025). *A commensal bacterial platform for intracellular drug delivery in TNBC*, Oral Presentation and Q&A, Paris Convention Centre, Paris, France (30 Oct 2025)

Zhong, J., et al. (2025). *A commensal bacterial platform for intracellular drug delivery in TNBC*, Oral Presentation, Conference of China iGEM Community & Global Bio-Developer Conference, Zhongguancun Exhibition Center, Beijing, China (7 Aug 2025)

Zhong, J. (2025). *A commensal bacterial platform for intracellular drug delivery in TNBC*, Oral Presentation, Internal Research Presentation, XJTLU School of Science, Suzhou, China (20 Oct 2025)

Additional Experience

Visiting Research Student, Qingdao Institute of Bioenergy and Bioprocess Technology, CAS (Jul 2025)

- Led technical discussions on restriction–modification systems, environmental sensing, and drug-release strategies with institute researchers; contributed to establishing an effective genetic transformation method for *Staphylococcus xylosus*.

Member, Suzhou Society for Biology (May 2025–present)

Class Representative & Student Community Representative, XJTLU (Sep 2023–Jun 2024)

Additional Skills

Technical Skills:

- **Experimental:** Mammalian cell culture, tumor spheroid culture, T-cell isolation, activation, transduction and expansion, viral packaging, HEK293F transfection; bacterial culture and transformation; PCR, Gibson and Golden Gate assembly; protein purification, Western blotting, ELISA; mouse tumor inoculation, tail-vein injection, tissue harvesting and dissection; flow cytometry, Transwell migration and LDH cytotoxicity assays; confocal microscopy, dynamic light scattering (DLS).
- **Computational & Image Analysis:** Python, Fiji, Imaris, FlowJo, PyMOL.
- **Visualization:** BioRender, Adobe Illustrator, Adobe Photoshop.

Languages:

- English: IELTS Academic 8.0 (Transcript↗)
- Chinese (Mandarin): Native

Extracurricular Activities

Sports

- Road Cycling: Road Race 47th, Individual Time Trial 35th, China U&C Student Cycling Challenge (May 2024).
- Basketball: Middle & High School Varsity Teams (2017–23); XJTLU School of Science Team (2023–25).
- Strength Training & Calisthenics.

Music

- Chinese Bamboo Flute (竹笛) and Chinese vertical bamboo flute (萧).